

GYZYLARBAT, Turkmenistan

WMO: 387630

Lat: 38.983N

Lon: 56.283E

Elev: 92

StdP: 100.22

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.8	-6.0	-14.1	1.1	-7.5	-10.7	1.5	-1.3	14.2	2.2	12.3	5.1	4.4	90	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	41.0	21.8	39.6	21.6	38.3	21.4	26.0	34.9	24.9	34.5	23.9	34.0	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.6	31.7	21.8	16.7	31.0	20.5	15.4	31.1	81.2	34.6	76.4	34.4	72.3	34.0	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.4	9.4	DB	-10.9	43.8	5.5	1.7	-14.9	45.0	-18.2	46.0	-21.3	46.9	-25.3	48.1	(4)
(5)			WB	-11.6	27.8	5.4	1.7	-15.5	29.0	-18.7	30.0	-21.7	31.0	-25.7	32.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	3.2	4.8	10.8	17.5	24.5	29.6	32.1	31.1	25.0	17.3	9.5	4.4	(6)
(7)		DBStd	11.18	4.82	5.28	4.49	4.66	3.56	3.11	2.54	2.76	3.56	4.59	4.94	4.32	(7)
(8)		HDD10.0	662	215	152	42	3	0	0	0	0	0	5	64	181	(8)
(9)		HDD18.3	1932	470	378	237	69	3	0	0	0	2	74	267	433	(9)
(10)		CDD10.0	3418	3	8	68	228	449	588	684	653	449	231	49	6	(10)
(11)		CDD18.3	1646	0	0	4	44	194	338	426	395	202	41	2	0	(11)
(12)		CDH23.3	22189	0	1	61	434	2059	4699	6496	5836	2228	356	18	0	(12)
(13)		CDH26.7	12902	0	0	17	158	954	2781	4162	3671	1055	100	4	0	(13)
(14)	Wind	WSAvg	3.8	3.8	4.1	4.3	4.4	3.8	4.1	3.7	3.5	3.8	3.6	3.5	3.8	(14)
(15)	Precipitation	PrecAvg	203	23	25	25	24	22	8	6	3	5	14	18	24	(15)
(16)		PrecMax	354	73	60	72	85	115	71	65	21	46	60	66	65	(16)
(17)		PrecMin	87	1	2	4	2	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	65	15	16	16	19	24	14	12	5	10	14	15	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	21.0	29.0	34.2	38.0	41.6	43.2	42.9	38.2	32.3	25.7	19.0	(19)
(20)			MCWB	9.7	10.9	14.8	18.6	20.6	21.8	23.1	22.5	20.9	18.6	14.7	10.2	(20)
(21)		2%	DB	14.2	16.8	24.4	30.8	35.5	39.7	41.2	40.7	35.6	29.3	20.9	15.3	(21)
(22)			MCWB	8.5	10.0	13.7	17.8	19.5	21.1	22.2	21.6	20.1	17.9	13.2	9.6	(22)
(23)		5%	DB	11.3	13.8	21.0	28.1	33.6	38.3	39.8	39.3	33.9	27.1	18.6	12.7	(23)
(24)			MCWB	7.5	8.9	12.8	16.6	19.2	20.9	21.9	21.4	19.7	16.7	12.3	8.6	(24)
(25)		10%	DB	9.2	11.7	18.2	25.6	31.7	36.7	38.5	37.8	32.1	25.0	16.5	10.4	(25)
(26)			MCWB	6.2	7.8	11.5	15.5	18.9	20.6	21.8	21.3	19.2	15.7	11.4	7.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	13.0	16.1	20.1	23.1	25.7	27.6	27.6	24.3	21.1	16.2	12.0	(27)
(28)			MCD B	16.1	19.1	24.7	30.7	32.6	35.8	36.0	35.7	33.1	28.3	22.4	17.3	(28)
(29)		2%	WB	9.0	10.5	14.4	18.5	21.5	24.1	26.1	25.9	22.8	19.2	14.5	10.1	(29)
(30)			MCD B	13.5	15.2	22.2	28.5	31.8	34.6	35.3	34.1	32.0	26.8	19.1	14.3	(30)
(31)		5%	WB	7.6	9.2	13.2	17.4	20.5	23.2	25.2	24.7	21.6	17.8	13.1	8.8	(31)
(32)			MCD B	11.0	13.3	20.0	26.3	30.6	33.5	34.9	34.3	30.8	25.3	17.7	12.3	(32)
(33)		10%	WB	6.4	8.0	12.0	16.3	19.6	22.3	24.2	23.6	20.3	16.6	11.8	7.6	(33)
(34)			MCD B	8.9	11.7	17.4	24.1	29.2	32.8	34.1	34.0	30.0	23.8	15.9	10.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	8.3	9.9	10.6	11.2	11.5	11.2	12.4	11.6	8.6	6.9	(35)	
(36)			MCD BR	11.5	12.6	15.2	14.7	14.0	13.8	13.0	14.0	14.6	13.9	12.0	11.2	(36)
(37)			MCWBR	6.8	7.2	6.9	6.0	5.5	5.4	5.2	5.7	7.1	6.8	6.1	6.9	(37)
(38)		5% WB	MCD BR	10.4	11.3	13.1	12.9	12.1	11.0	10.4	10.6	11.5	11.8	9.9	10.1	(38)
(39)			MCWBR	6.5	6.9	6.7	5.9	5.7	5.6	5.2	5.7	6.9	6.6	5.5	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.403	0.453	0.509	0.575	0.560	0.559	0.578	0.530	0.505	0.500	0.439	0.409	(40)	
(41)		taud	2.049	1.920	1.816	1.720	1.784	1.795	1.760	1.847	1.877	1.869	1.995	2.034	(41)	
(42)		Ebn at Noon	724	738	740	721	744	745	725	749	735	678	673	679	(42)	
(43)		Edh at Noon	122	159	195	228	218	216	222	198	181	165	128	116	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.11	2.92	4.12	5.29	6.66	7.68	7.36	6.84	5.61	3.89	2.60	1.91	(44)	
(45)		RadStd	0.19	0.33	0.28	0.53	0.33	0.22	0.28	0.33	0.19	0.22	0.19	0.21	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page